

**ARBEITSGEMEINSCHAFT:
NEW HOMOLOGICAL METHODS IN TORIC GEOMETRY
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The goal of this Arbeitsgemeinschaft is to highlight several recent generalizations to toric varieties of landmark theorems about the derived category of projective space. Specifically, a theorem of Beilinson [Bei78] gives a resolution of the diagonal sheaf on $\mathbb{P}^n \times \mathbb{P}^n$, leading to a full strong exceptional collection for the derived category of $\mathbb{P}^n$ given by the line bundles $\mathcal{O}(-n), \dots, \mathcal{O}(-1), \mathcal{O}$, the *Beilinson collection*. A closely related theorem due to Bernstein-Gel'fand-Gel'fand [BGG78] states that the derived category of $\mathbb{P}^n$ is equivalent to the stable category of modules over an exterior algebra. These theorems launched the study of derived categories in algebraic geometry.

A breakthrough 2024 theorem of Hanlon-Hicks-Lazarev [HHL24] generalized Beilinson's resolution of the diagonal to toric varieties, and subsequent work of Ballard et al. [BBB⁺24] applied Hanlon-Hicks-Lazarev's theorem to give a toric version of Beilinson's collection. Around the same time, Brown-Erman [BE26] proved a toric analogue of Bernstein-Gel'fand-Gel'fand's theorem. Proving—and even stating—these results requires a substantial departure from the case of $\mathbb{P}^n$, involving entirely new homological and categorical constructions. The aim of this week is to understand these three theorems.

We will assume some familiarity with algebraic geometry (varieties and sheaf cohomology) and with homological algebra (free resolutions and derived functors like Ext and Tor). We will also assume a small amount of knowledge toric varieties, e.g. [CLS11, §3]. Participants may find it useful to read about derived categories and the aforementioned theorems of Beilinson and Bernstein-Gel'fand-Gel'fand in, for example, [Că105, §3] and [EFS03, §2]. Additional knowledge of toric varieties, such as [CLS11, §5, §6, §9, §14] would be helpful, but not assumed.

The Hirzebruch surface of type 3, which we denote as $\mathcal{H}_3$, will serve as a running example through the week, and participants should be familiar with this example. See [CLS11, Example 3.1.16]. In addition, the organizers will prepare a demo package in `Macaulay2` so that the speakers (and participants) can more easily compute explicit examples to illustrate the relevant results and constructions.

MONDAY: BACKGROUND ON THE DERIVED CATEGORY OF $\mathbb{P}^n$ AND TORIC VARIETIES

1. DERIVED CATEGORY OF $\mathbb{P}^n$, I

Briefly define the bounded derived category of a variety, and briefly explain that it is an example of a triangulated category. Define the notion of a Fourier-Mukai transform between derived categories of varieties. Explain why the Fourier-Mukai transform with kernel given by the diagonal sheaf is isomorphic to the identity functor. Discuss Beilinson's resolution of the diagonal sheaf on $\mathbb{P}^n \times \mathbb{P}^n$.

2. DERIVED CATEGORY OF $\mathbb{P}^n$, II

Define semiorthogonal decompositions and (full, strong) exceptional collections for triangulated categories. As a key example, use Beilinson's resolution of the diagonal to prove that the Beilinson collection gives a full strong exceptional collection for $D^b(\mathbb{P}^n)$ [Că105, §3]. Discuss how this implies

that the Grothendieck group $K_0(\mathbb{P}^n)$ is isomorphic to $\mathbb{Z}^{n+1}$. This theorem launched a search for exceptional collections and semiorthogonal decompositions for other varieties. As time permits, provide a brief overview of some results in this vein, such as Kapranov's exceptional collections for smooth quadrics and Grassmanians [Kap88], or Orlov's blowup formula [Huy06, Proposition 11.18] and projective bundle formula [Huy06, Corollary 8.36]. See [Kuz14] for a survey on this topic.

3. DERIVED CATEGORY OF $\mathbb{P}^n$, III

Describe the Bernstein-Gel'fand-Gel'fand (BGG) correspondence, which gives an equivalence of bounded derived categories between a graded polynomial ring S and its Koszul dual exterior algebra E . Describe the functors between these two categories, following [EFS03, §2], and include some explicit examples. Illustrate that these functors are inverse equivalences by evaluating their compositions at the residue field. (A detailed proof that these functors are inverse equivalences is too much for one lecture.) Letting $D_{\mathfrak{m}}^b(S)$ denote the subcategory of $D^b(S)$ given by complexes with homology supported in the homogeneous maximal ideal $\mathfrak{m}$, explain that the BGG correspondence identifies $D_{\mathfrak{m}}^b(S)$ with perfect complexes in $D^b(E)$. Briefly discuss Verdier quotients of triangulated categories, and note that $D^b(S)/D_{\mathfrak{m}}^b(S)$ is equivalent to $D^b(\mathbb{P}^n)$. Conclude that the BGG correspondence identifies $D^b(\mathbb{P}^n)$ with the singularity category of E , i.e. $D^b(E)$ modulo perfect complexes.

4. TORIC VARIETIES AND COX RINGS, I

Review how to associate a normal toric variety to a fan [CLS11, §3.1]. Given a fan Σ and the associated toric variety X_{Σ} , explain how the rays in Σ determine divisors on X_{Σ} via the Orbit-Cone Correspondence [CLS11, §4.1]. Define the Cox ring (also called the homogeneous coordinate ring) S of X_{Σ} , its grading by the divisor class group $\text{Cl}(X_{\Sigma})$, and its irrelevant ideal B . Explain how the variety X_{Σ} arises as the quotient of $\text{Spec}(S) \setminus V(B)$ by a torus action. Potential references for Cox rings are [CLS11, §5.1] or Cox's original paper [Cox95]. Include running examples throughout, especially projective space, products of projective spaces, weighted projective spaces, and Hirzebruch surfaces $\mathcal{H}_i$ for $i \geq 1$.

5. TORIC VARIETIES AND COX RINGS, II

Describe the Toric Ideal-Variety Correspondence [CLS11, Proposition 5.2.7]. Demonstrate by example that the Cox rings of non-isomorphic toric varieties can be isomorphic as graded rings. Describe how graded modules over the Cox ring induce sheaves on toric varieties [CLS11, §5.3]. Explain that when X_{Σ} is a smooth toric variety associated to a fan Σ with Cox ring S , there is an equivalence $D^b(X_{\Sigma}) \simeq D^b(S)/D_B^b(S)$, where $D_B^b(S)$ denotes the subcategory of complexes with support in $V(B)$: this follows essentially from [CLS11, Propositions 5.3.9 and 5.3.10]. Explain how this can fail when X_{Σ} is not smooth (see e.g. [CLS11, Exercise 5.3.5]). Finally, state the Demazure vanishing theorem [CLS11, Theorem 9.2.3].

TUESDAY: THE HANLON-HICKS-LAZAREV RESOLUTION

6. THE BONDAL-THOMSEN COLLECTION.

Define the Bondal-Thomsen collection, which plays the role of the Beilinson collection in the toric setting. A reference is [HHL24, Section 2.3] (where it is referred to as simply the Thomsen collection). This reference handles the generality of toric stacks [?GS], but it is preferable here to stick to the case of toric varieties (i.e. the case where the map β in [HHL24, §2.3] is the identity map). Compute several examples, e.g. $\mathbb{P}^n$ and Hirzebruch surfaces. If time allows, weighted projective spaces and the varieties in the Atiyah flop are also valuable examples. Discuss [HHL24,

Corollary 5.5], which identifies the Bondal-Thomsen collection with summands of a Frobenius pushforward. Show that the Bondal-Thomsen collection for $\mathcal{H}_3$ has 6 elements but that, unlike what happens for $\mathbb{P}^n$, these do not form an exceptional collection.

7. CONSTRUCTING THE HANLON-HICKS-LAZAREV RESOLUTION

Explain the construction of the Hanlon-Hicks-Lazarev resolution, following the exposition in [BH25]. Once again, the construction is presented at the level of toric stacks, but it is preferable to focus on varieties. Work through several examples: a point in $\mathbb{P}^2$; the diagonal in $\mathbb{P}^2$ (compare with Beilinson's resolution); the diagonal of a Hirzebruch surface; the twisted cubic in $\mathbb{P}^3$; more as time permits. Remark that resolutions of the diagonal with properties of the HHL resolution (e.g. with length given by the dimension of the variety and terms given by sums of line bundles) had been constructed previously in several special cases, see e.g. [BE26, BPS01, BS24, CK08].

There is Macaulay2 code for computing Hanlon-Hicks-Lazarev resolutions, which will be provided in a demo package. This lecturer might consider including an explicit Macaulay2 demonstration as part of this lecture.

8. MORE EXAMPLES AND FEATURES OF THE HANLON-HICKS-LAZAREV RESOLUTION

Discuss additional examples of HHL resolutions, and emphasize that the HHL resolution of the diagonal for a toric variety X depends only on the rays of the fan determining the X .

9. PROOF OF THE HANLON-HICKS-LAZAREV THEOREM

Prove the Hanlon-Hicks-Lazarev theorem [HHL24, Theorem A], following the simplified approach introduced in the recent preprints [BCS+25] and [BH25]. Mention the connection to the integral closure of the coordinate ring and thus to [BE24], as proven in [BCS+25, §4].

WEDNESDAY: APPLICATIONS AND THE COX CATEGORY

10. APPLICATIONS OF THE HANLON-HICKS-LAZAREV THEOREM

Discuss four main applications of the Hanlon-Hicks-Lazarev Theorem:

- (1) A smooth toric stack X admits a resolution of the diagonal of length $\dim(X)$ whose terms are sums of line bundles. This gives the desired toric generalization of Beilinson's resolution of the diagonal over $\mathbb{P}^n$.
- (2) Berksech-Erman-Smith's conjecture [BES20, Question 6.5] on short virtual resolutions: see [HHL24, Corollary C].
- (3) The Bondal-Thomsen collection generates the derived category [HHL24, Corollary D]. A different proof of this fact appears in [FH23, Corollary 3.5].
- (4) The toric case of Orlov's Conjecture [Orl09, Conjecture 10]: see [HHL24, Corollary E]. Again, a different proof can be found in [FH23, Theorem 1.2].

Briefly discuss the history of partial results concerning (1) - (4), as outlined in [HHL24, Section 1].

11. THE SECONDARY FAN

Discuss the secondary fan of a toric variety X , following the terse exposition in [BBB⁺24, §2.1]. The key point to describe here is that the (finitely many) maximal cones of the secondary fan parameterize the simplicial, projective toric varieties with the same Cox ring as X —i.e. the projective simplicial fans with the same rays as X —as well as the projective simplicial fans whose rays are a subset of the rays of X .

12. THE COX CATEGORY, I: DEFINITION

Briefly discuss King's conjecture and its history. Define the Cox category, establish its basic properties (as in [BBB⁺24, §3]), and discuss some examples. State the revised King's Conjecture.

Evening discussion: toric stacks. Why do toric stacks arise in this story, and how does one work with them in practice?

THURSDAY: PROOF OF THE REVISED KING CONJECTURE

13. THE COX CATEGORY, II: FOURIER-MUKAI TRANSFORMS

Given a smooth projective toric variety X , compute examples of Fourier-Mukai transformations of line bundles on X to the toric varieties associated to the maximal chambers of its secondary fan. For instance, show that the Fourier-Mukai transform $D(\mathcal{H}_3) \rightarrow D(\mathbb{P}(1, 1, 3))$ sends the line bundle $\mathcal{O}(-1, 0)$ to $\mathcal{O}(-1)$. These examples will provide motivation for the Θ -Transform Lemma ([BBB⁺24, Lemma 1.6]), which is the key technical result in [BBB⁺24].

14. PROOF OF REVISED KING'S CONJECTURE, I: OUTLINE OF MAIN IDEAS

Prove the revised King's conjecture when there is only one chamber of the secondary fan. Then state the Θ -Transform Lemma and demonstrate via several examples, like [BBB⁺24, §4.2]. Outline how to obtain the Revised King Conjecture from the Θ -Transform Lemma ([BBB⁺24, Lemma 1.6]).

15. PROOF OF REVISED KING'S CONJECTURE, II: EXCEPTIONALITY

Prove the Θ -Transform Lemma, following [BBB⁺24, §4.3]. Illustrate the key steps through the lens of the Bondal-Thomsen elements on the Hirzebruch surface $\mathcal{H}_3$.

16. PROOF OF REVISED KING'S CONJECTURE, III: CONCLUSION

Finish the proof of the Revised King's Conjecture, following [BBB⁺24, §4.3 and §5].

FRIDAY: APPLICATIONS OF COX CATEGORIES/TATE RESOLUTIONS

17. APPLICATIONS OF COX CATEGORIES

Discuss some applications of [BBB⁺24, Theorem A], potentially including [BBB⁺24, Proposition 1.3, Theorems 1.4, 1.5], at the discretion of the lecturer.

18. TATE RESOLUTIONS OVER $\mathbb{P}^n$

Recall from Lecture 3 that $D^b(\mathbb{P}^n)$ is equivalent to the singularity category of an exterior algebra E . Explain how the singularity category of E , in turn, is equivalent to the homotopy category $K(E)$ of exact complexes of finitely generated graded free E -modules [Buc21, Theorem 4.4.1]. The BGG correspondence for $\mathbb{P}^n$ may therefore be realized as an equivalence $T: D^b(\mathbb{P}^n) \xrightarrow{\sim} K(E)$. The equivalence T is the Tate resolution functor described in [EFS03, §4]. Explain the formula for this functor, evaluated just on coherent sheaves (not complexes of such), and compute some explicit examples. Displaying Macaulay2 computations using the Macaulay2 package BGG could be useful. Discuss the calculation of the terms of the Tate resolution in terms of sheaf cohomology [EFS03, Theorem 4.4.1]. Explain also the application to effectively computing Beilinson monads in [EFS03, Theorem 6.1].

19. TATE RESOLUTIONS ON TORIC VARIETIES

Briefly explain the multigraded BGG correspondence, as described in [BE26, §2]. Then state and prove [BE26, Theorem 3.3], which gives the toric analogue of Tate resolutions on toric varieties. Compute some examples over weighted projective stacks as in [BE26, §3.4]. Display some example computations using the MultigradedBGG package in Macaulay2 [BBG⁺25].

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